To establish a sharp asymptotic upper bound on the predictive risk of the transfer learning estimator $\hat{\omega}_{\mathrm{TL}}^{(t)}$ and determine the optimal penalty parameter λ_ω, we synthesize the risk decomposition framework of Gu, Han, and Duan (2025) with the high-dimensional deterministic equivalent calculus of Dobriban and Wager (2018).

Setup and Regularity Conditions

We supplement the provided assumptions with two standard regularity conditions to ensure tractability of the deterministic equivalents and rigorous bounds on the cross-terms.

1. **Condition 1 (Isotropic Design):** We assume the feature covariance is $\Sigma = I_p$. This simplifies the projection matrix to $\Phi^{(t)} = [I_{p_t}, 0_{p_t \times (p-p_t)}]$, mapping the full coefficients to the observed dimensions $\omega^{(t)} = \beta^{(t)}_{\mathcal{R}(t)}$.
2. **Condition 2 (Independence of Source Noise):** The source estimation error $\Delta_s = \hat{\beta}^{(s)} - \beta^{(s)}$ is assumed to be independent of the target regression coefficient $\beta^{(t)}$ and the target-site data.

Under Condition 1, the projection matrix operates purely by truncation, avoiding the complex correlation structure of the unobserved features.

Part 1: Predictive Risk Decomposition

At the target site, the true linear model is $Y^{(t)} = X^{(t)}\beta^{(t)} + \varepsilon$. We partition the features into the observed set $\mathcal{R}(t)$ and the unobserved set $[\mathcal{R}(t)]^c$:

$$Y^{(t)} = X^{(t)}_{\mathcal{R}(t)}\beta^{(t)}_{\mathcal{R}(t)} + X^{(t)}_{[\mathcal{R}(t)]^c}\beta^{(t)}_{[\mathcal{R}(t)]^c} + \varepsilon.$$

Let $\omega^{(t)} = \beta^{(t)}_{\mathcal{R}(t)}$. Because $X^{(t)}$ has independent standard normal columns under $\Sigma = I_p$, the unobserved features act as isotropic noise conditionally on $X^{(t)}_{\mathcal{R}(t)}$. We can define an effective noise vector $\tilde{E}^{(t)} = X^{(t)}_{[\mathcal{R}(t)]^c}\beta^{(t)}_{[\mathcal{R}(t)]^c} + \varepsilon$, yielding the working model:

$$Y^{(t)} = X^{(t)}_{\mathcal{R}(t)}\omega^{(t)} + \tilde{E}^{(t)}.$$

The conditional variance of the effective noise, taking the expectation over the random coefficients and ε, is:

$$\tau_n^2 := \mathrm{Var}(\tilde{E}_i^{(t)}) = \sigma^2 + \mathbb{E}\left[\|\beta^{(t)}_{[\mathcal{R}(t)]^c}\|_2^2\right] = \sigma^2 + \frac{\alpha_t^2}{p}(p - p_t).$$

As $n_t, p, p_t \to \infty$ with $p/n_t \to \gamma$ and $p_t/n_t \to \eta$, this variance converges to a deterministic limit:

$$\tau^2 := \lim_{n_t \to \infty} \tau_n^2 = \sigma^2 + \alpha_t^2 \left(1 - \frac{\eta}{\gamma}\right).$$

For a new independent test observation, the predictive risk decomposes into the irreducible effective noise variance and the estimation error:

$$r^{\text{pred}}\big(\hat{\omega}_{\text{TL}}^{(t)}\big) = \tau^2 + \mathbb{E}\left[\big\|\hat{\omega}_{\text{TL}}^{(t)} - \omega^{(t)}\big\|_2^2\right].$$

Part 2: Bias and Variance of the Estimator

The transfer learning estimator is the solution to a Ridge-type penalized regression centered at $\hat{\omega}_s^{(t)} = \Phi^{(t)}\hat{\beta}^{(s)}$. Taking the gradient of the objective function and setting it to zero yields the closed-form solution:

$$\hat{\omega}_{\text{TL}}^{(t)} = (\hat{\Sigma}_t + \lambda_\omega I_{p_t})^{-1}\left(\frac{1}{n_t}\big(X_{\mathcal{R}(t)}^{(t)}\big)^\top Y^{(t)} + \lambda_\omega \hat{\omega}_s^{(t)}\right),$$

where $\hat{\Sigma}_t = \frac{1}{n_t}\big(X_{\mathcal{R}(t)}^{(t)}\big)^\top X_{\mathcal{R}(t)}^{(t)}$ is the sample covariance matrix of the observed target features.

Let $S_\lambda = (\hat{\Sigma}_t + \lambda_\omega I_{p_t})^{-1}$. Substituting $Y^{(t)} = X_{\mathcal{R}(t)}^{(t)}\omega^{(t)} + \tilde{E}^{(t)}$ into the estimator and using the identity $S_\lambda \hat{\Sigma}_t - I_{p_t} = -\lambda_\omega S_\lambda$, the estimation error $\Delta = \hat{\omega}_{\text{TL}}^{(t)} - \omega^{(t)}$ becomes:

$$\Delta = S_\lambda\left(\frac{1}{n_t}\big(X_{\mathcal{R}(t)}^{(t)}\big)^\top \tilde{E}^{(t)} + \lambda_\omega\big(\hat{\omega}_s^{(t)} - \omega^{(t)}\big)\right).$$

Because $\tilde{E}^{(t)}$ is mean-zero and independent of the transferred prior, the expected squared error splits additively into a variance term V (from the target-site effective noise) and a bias term B (from the transfer prior):

$$\mathbb{E}\left[\|\Delta\|_2^2\right] = \underbrace{\frac{\tau^2}{n_t}\mathbb{E}\left[\operatorname{Tr}(S_\lambda^2 \hat{\Sigma}_t)\right]}_{\text{Variance } V} + \underbrace{\lambda_\omega^2 \mathbb{E}\left[\big(\hat{\omega}_s^{(t)} - \omega^{(t)}\big)^\top S_\lambda^2 \big(\hat{\omega}_s^{(t)} - \omega^{(t)}\big)\right]}_{\text{Bias } B}.$$

Bounding the Transfer Bias

We must bound the covariance of the transfer prior error $H = \hat{\omega}_s^{(t)} - \omega^{(t)}$. Using $\Phi^{(t)}$, we can write:

$$H = \Phi^{(t)}(\hat{\beta}^{(s)} - \beta^{(s)}) + \Phi^{(t)}(\beta^{(s)} - \beta^{(t)}).$$

By Condition 2, the source estimation error and the coefficient drift are independent. We bound the second moment matrix:

$$\mathbb{E}[HH^\top] = \Phi^{(t)}\mathbb{E}\left[(\hat{\beta}^{(s)} - \beta^{(s)})(\hat{\beta}^{(s)} - \beta^{(s)})^\top\right](\Phi^{(t)})^\top + \Phi^{(t)}\mathbb{E}\left[(\beta^{(s)} - \beta^{(t)})(\beta^{(s)} - \beta^{(t)})^\top\right](\Phi^{(t)})^\top.$$

From the Source Estimation Error Assumption, the largest eigenvalue of the first term is bounded by C^U/p. For the second term, the Random Regression Coefficients Assumption yields:

$$\mathbb{E}\left[(\beta^{(s)}-\beta^{(t)})(\beta^{(s)}-\beta^{(t)})^\top\right]=\frac{\alpha_s^2+\alpha_t^2-2\rho\alpha_s\alpha_t}{p}I_p.$$

Let $C_{\text{total}}=C^U+\alpha_s^2+\alpha_t^2-2\rho\alpha_s\alpha_t$. Since $\Phi^{(t)}(\Phi^{(t)})^\top=I_{p_t}$, we establish the uniform upper bound:

$$\mathbb{E}[HH^\top]\preceq\frac{C_{\text{total}}}{p}I_{p_t}.$$

Consequently, the bias term is bounded by:

$$B\leq\lambda_\omega^2\frac{C_{\text{total}}}{p}\mathbb{E}\left[\text{Tr}(S_\lambda^2)\right].$$

Part 3: Asymptotic Deterministic Equivalents

We now apply the limiting trace calculations from Dobriban and Wager (2018). As $n_t, p_t\to\infty$ with $p_t/n_t\to\eta$, the empirical spectral distribution of $\hat{\Sigma}_t$ converges almost surely to the Marchenko-Pastur (MP) law.

Let $m(-\lambda_\omega)$ be the Stieltjes transform of the MP law evaluated at $-\lambda_\omega$, which is the unique positive solution to:

$$m=\frac{1}{\lambda_\omega+\frac{\eta}{1+m}}.$$

By the properties of Stieltjes transforms, the normalized traces converge as follows:

$$\frac{1}{p_t}\text{Tr}(S_\lambda)\xrightarrow{a.s.}m(-\lambda_\omega):=m.$$

Taking the derivative with respect to λ_ω gives the limit for $\frac{1}{p_t}\text{Tr}(S_\lambda^2)$:

$$\frac{1}{p_t}\text{Tr}(S_\lambda^2)\xrightarrow{a.s.}m'(-\lambda_\omega):=m'=\frac{m^2(1+m)^2}{(1+m)^2-\eta m^2}.$$

Using the identity $S_\lambda^2\hat{\Sigma}_t=S_\lambda-\lambda_\omega S_\lambda^2$, we evaluate the variance trace limit:

$$\frac{1}{n_t}\text{Tr}(S_\lambda^2\hat{\Sigma}_t)=\frac{p_t}{n_t}\frac{1}{p_t}\text{Tr}(S_\lambda^2\hat{\Sigma}_t)\xrightarrow{a.s.}\eta(m-\lambda_\omega m').$$

Substituting these deterministic equivalents back into the bounding expressions for V and B, the asymptotic predictive risk is bounded by:

$$\bar{R}(\lambda_\omega)=\tau^2+\eta\tau^2(m-\lambda_\omega m')+\lambda_\omega^2\frac{\eta}{\gamma}C_{\text{total}}m'.$$

Part 4: Optimal Regularization and Final Bound

To find the optimal regularization parameter λ_ω^*, we minimize the asymptotic risk upper bound $\bar{R}(\lambda_\omega)$ with respect to λ_ω. Note that m and m' are functions of λ_ω, but we can optimize directly over the spectral integral representation:

$$\bar{R}(\lambda_\omega) = \tau^2 + \int \frac{\eta\tau^2 x + \frac{\eta}{\gamma} C_{\text{total}} \lambda_\omega^2}{(x + \lambda_\omega)^2} dF(x),$$

where $F(x)$ is the MP distribution. Taking the derivative with respect to λ_ω and setting it to zero:

$$\frac{\partial \bar{R}}{\partial \lambda_\omega} = \int \frac{2x\left(\frac{\eta}{\gamma} C_{\text{total}} \lambda_\omega - \eta\tau^2\right)}{(x + \lambda_\omega)^3} dF(x) = 0.$$

This equation is satisfied for all x when the numerator vanishes. Solving for λ_ω yields the theoretically optimal transfer penalty:

$$\lambda_\omega^* = \frac{\gamma\tau^2}{C_{\text{total}}} = \frac{\gamma\left(\sigma^2 + \alpha_t^2(1 - \eta/\gamma)\right)}{C^U + \alpha_s^2 + \alpha_t^2 - 2\rho\alpha_s\alpha_t}.$$

Substituting λ_ω^* back into the risk expression creates a massive simplification. Notice that at the optimal λ_ω^*, we have $\frac{\eta}{\gamma} C_{\text{total}} (\lambda_\omega^*)^2 = \eta\tau^2 \lambda_\omega^*$. The bias and the second half of the variance term perfectly collapse:

$$\bar{R}(\lambda_\omega^*) = \tau^2 + \eta\tau^2 m - \eta\tau^2 \lambda_\omega^* m' + \eta\tau^2 \lambda_\omega^* m' = \tau^2 + \eta\tau^2 m(-\lambda_\omega^*).$$

Final Conclusion: The predictive risk of the transfer learning estimator $\hat{\omega}_{\text{TL}}^{(t)}$ is asymptotically bounded above by:

$$\limsup_{n_t, p \to \infty} r^{\text{pred}}\big(\hat{\omega}_{\text{TL}}^{(t)}\big) \leq \tau^2\big(1 + \eta \cdot m(-\lambda_\omega^*)\big),$$

where $\tau^2 = \sigma^2 + \alpha_t^2(1 - \eta/\gamma)$, $\lambda_\omega^* = \frac{\gamma\tau^2}{C^U + \alpha_s^2 + \alpha_t^2 - 2\rho\alpha_s\alpha_t}$, and $m(-\lambda_\omega^*)$ is the unique positive root of $m = \left(\lambda_\omega^* + \frac{\eta}{1+m}\right)^{-1}$. $\square$